ThermoDT: Learning Transferable Indoor Thermal Field Operators from Sparse Wireless Sensor Networks

Nueraili Kuerbanjiang
Carnegie Mellon University
Civil and Environmental Engineering
Pittsburgh, Pennsylvania, USA
nkuerban@andrew.cmu.edu

David Delgado Ferrufino
Carnegie Mellon University
Civil and Environmental Engineering
Pittsburgh, Pennsylvania, USA
dadelgad@andrew.cmu.edu

Justin Oh
Carnegie Mellon University
Civil and Environmental Engineering
Pittsburgh, Pennsylvania, USA
jhoh@andrew.cmu.edu

Mario Bergés
Carnegie Mellon University
Civil and Environmental Engineering
Pittsburgh, Pennsylvania, USA
marioberges@cmu.edu

Abstract

Indoor thermal environments are spatially heterogeneous: temperature and humidity vary across a room because of airflow, equipment, occupancy, and localized heat sources, yet most building control systems still rely on a single thermostat or a handful of point measurements. We present **ThermoDT**, a neural operator-based thermal digital twin that reconstructs continuous 2D temperature and humidity fields from sparse wireless sensor networks. Our final model is a Deep Operator Network (DeepONet) whose branch network encodes the multi-sensor thermal state together with an explicit observed/missing mask, while the trunk network maps spatial query coordinates to a learned basis, enabling field evaluation at arbitrary room locations. The model is trained exclusively on real IoT data from a public dataset containing two indoor spaces (Lab 1 and Lab 2), each instrumented with 13 ceiling-mounted wireless sensors. On Lab 1, the model reconstructs temperature and humidity robustly under severe masking and retains strong performance with only a small number of active sensors. Exhaustive subset analysis shows that strategically chosen 2-sensor and 3-sensor subsets preserve most of the full-sensor performance. More importantly, a model trained on Lab 1 transfers meaningfully to Lab 2 zero-shot and improves substantially under lightweight adaptation. These results suggest that the learned operator captures transferable indoor thermo-hygrometric structure rather than memorizing a single room. These reconstructed temperature and humidity fields are then used to support spatial thermal comfort assessment at arbitrary room locations, making the digital twin directly useful for comfort-aware building analysis.

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Supervised learning*.

Keywords

thermal digital twin, neural operators, DeepONet, indoor sensing, sparse sensor networks, transfer learning, physics-informed learning, smart buildings

Unpublished working draft. Not for distribution.

12-770 Autonomous Sustainable Buildings: From Theory to Practice, Pittsburgh, PA USA

ACM Reference Format:
Nueraili Kuerbanjiang, David Delgado Ferrufino, Justin Oh, and Mario Bergés. 2026. ThermoDT: Learning Transferable Indoor Thermal Field Operators from Sparse Wireless Sensor Networks. In *Proceedings of CMU 12-770 Autonomous Sustainable Buildings: From Theory to Practice (12-770 Autonomous Sustainable Buildings: From Theory to Practice)*. ACM, New York, NY, USA, 7 pages.

Unpublished working draft. Not for distribution.

1 Introduction

Smart building systems require accurate, spatially resolved knowledge of indoor thermal conditions to optimize occupant comfort and energy efficiency. In practice, however, buildings are often monitored by a single thermostat or a small number of point sensors. This sparse instrumentation cannot capture room-scale thermal heterogeneity caused by HVAC discharge patterns, equipment loads, occupancy, solar gains, and local airflow.

Computational fluid dynamics (CFD) can, in principle, resolve these spatial gradients, but its computational cost makes real-time deployment difficult. Data-driven models offer a faster alternative, yet much of the prior literature either focuses on synthetic data or evaluates models only within the same environment used for training. For real building applications, two questions remain especially important:

(1) Can a model reconstruct a continuous indoor thermal field from sparse, irregular sensor measurements under realistic missing-sensor conditions?
(2) Can a model trained in one room generalize to a different room with different geometry and thermal structure?

This project addresses both questions using a public real-world IoT dataset collected in two indoor laboratories. The project has now converged on a clear core story. Rather than pursuing a broad set of loosely connected model variants, we center the final paper on one selected model, its sparse-sensor reconstruction capability, and its cross-room transfer behavior.

Our current contributions are:

- **ThermoDT**: a DeepONet-based thermal digital twin that reconstructs continuous 2D temperature and humidity fields from sparse, masked sensor inputs.

- **Robust sparse reconstruction**: on Lab 1, the model degrades gracefully under masking and retains strong performance even with very small active sensor sets.
- **Minimal sensor set analysis**: exhaustive subset search identifies strategically placed 2-sensor and 3-sensor subsets that preserve most of the full-sensor accuracy.
- **Cross-room transfer**: a model trained on Lab 1 transfers meaningfully to Lab 2 zero-shot, improves further with head-only adaptation, and reaches near-reference performance under full fine-tuning.
- **Thermal comfort mapping**: the reconstructed thermo-hygrometric field is used to generate spatial thermal comfort estimates at arbitrary room coordinates, linking the digital twin to a practical building-level outcome.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset and problem formulation. Section 4 presents the final model and key design choices. Section 5 summarizes the current results. Section 6 outlines the remaining experimental plan. Section 7 gives the week-by-week plan for the remainder of the project.

2 Related Work

2.1 Indoor Thermal Modelling from Sensor Data

[Placeholder: survey prior work on indoor temperature and humidity modelling from sparse sensor data, including interpolation-based virtual sensing, standard neural networks, and sequence models. The key point to emphasize is that most work predicts at fixed locations or is evaluated only within the same room used for training.]

2.2 Neural Operators and DeepONet

DeepONet [2] learns mappings between functions through two coupled subnetworks: a branch network that encodes the input function and a trunk network that encodes the query domain. This is particularly attractive for thermal digital twins because it enables continuous field reconstruction at arbitrary spatial coordinates rather than only predicting at a fixed set of sensor locations.

2.3 Transfer Learning in Smart Buildings

[Placeholder: brief survey of transfer learning and domain adaptation in building energy, HVAC, or indoor environmental modelling. The main gap to highlight is that cross-room transfer of a continuous spatial thermal field operator on real sensor data appears to be relatively unexplored.]

2.4 Explored but Secondary Directions

We also explored temporal and physics-informed variants of the model. These directions were useful for understanding the problem and establishing design choices, but they are not the primary focus of the final report. At present, the strongest and most coherent story is based on the final sparse-reconstruction model and its cross-room transfer behavior.

3 Data and Problem Formulation

3.1 Dataset

All experiments use the Botero-Valencia et al. dataset [1], a public IoT dataset collected in two real indoor environments (Lab 1 and Lab 2). Each room contains 13 previously calibrated wireless sensors suspended from the ceiling at known coordinates. The dataset includes temperature, relative humidity, and RSSI sampled at approximately 10-second intervals.

For our modelling pipeline, we resample the temperature and humidity channels to 5-minute resolution. The primary within-room experiments are performed on Lab 1. Lab 2 is used for transfer evaluation.

[Figure placeholder: Lab 1 floor plan with sensor positions overlaid. The figure should show room geometry and the 13 sensor locations used by the model.]

Figure 1: Lab 1 floor plan and sensor positions. The 13 wireless sensors define the sparse measurement geometry used by the operator network. Their coordinates are used both as input metadata and as target query locations during reconstruction.

[Figure placeholder: Lab 2 floor plan with sensor positions overlaid, showing its different aspect ratio and relative to Lab 1.]

Figure 2: Lab 2 floor plan and sensor positions. Lab 2 differs from Lab 1 in geometry and thermal structure, making it a meaningful transfer target rather than a trivial repetition of the same environment.

3.2 Problem Formulation

Let $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{N}$ denote the set of sensor coordinates, with $N = 13$. At each time step t, the observed room state is a sparse set of temperature and humidity values measured at some subset of active sensors. The remaining sensors may be intentionally masked for training and evaluation or unavailable in a deployment scenario.

Our goal is to learn an operator

$$\hat{\mathbf{z}}(x, y; t) = \mathcal{G}(\mathbf{u}(t), \mathbf{m}(t); x, y) \tag{1}$$

where:

- $\mathbf{u}(t)$ is the vector of sensor readings at time t,
- $\mathbf{m}(t) \in \{0, 1\}^N$ is the observed/missing mask,
- (x, y) is any spatial query coordinate in the room,
- $\hat{\mathbf{z}}(x, y; t)$ is the predicted 2-vector containing temperature and humidity.

The operator should reconstruct values both at sensor coordinates and at intermediate room locations, including coordinates for sensors that have been masked from the input.

Unpublished working draft. Not for distribution.

4 Methodology

4.1 Final Model Selection

After exploring multiple alternatives, including wider and deeper MLPs, a short-history LSTM branch, and other extensions, we selected the **v0.5.3** DeepONet configuration as the final model for the core study. It offered the best overall trade-off between accuracy, stability, and architectural simplicity.

4.2 Data Loading and Preprocessing

The final pipeline uses the resampled Lab 1 temperature and humidity files at 5-minute resolution. Each time step provides a full 13-sensor snapshot for both variables. Sensor coordinates are stored separately and used as the geometric support of the operator.

The final model uses:

- temperature and humidity only,
- 13 fixed sensor identities,
- one shared timestamp stream,
- one row per time step after resampling.

All variables are loaded into a consistent time-indexed table before splitting.

4.3 Normalization

Temperature and humidity are normalized independently using statistics computed from the training split only. These normalization parameters are then reused for validation, test, and transfer evaluation. This avoids leakage from future periods or from Lab 2.

Query coordinates are normalized by room width and height so that the trunk network operates in a consistent spatial domain.

4.4 Blocked Train/Validation/Test Split

For the main within-room experiments, we use a blocked temporal split rather than a random pointwise split. This is important because indoor thermal data are strongly autocorrelated in time. The later transfer experiments adopt a roughly 80/10/10 blocked strategy for Lab 1, and a separate blocked adaptation/test protocol for Lab 2.

4.5 DeepONet Architecture

The final model is a two-output DeepONet with one branch network and one trunk network.

Branch network. The branch input concatenates:

- 13 temperature values,
- 13 humidity values,
- a 13-dimensional observed/missing binary mask.

This gives a 39-dimensional input vector.

The branch is a 6-layer MLP with hidden width 128 and `tanh` activations.

Trunk network. The trunk input is the normalized query coordinate $(x, y) \in [0, 1]^2$. It is also a 6-layer MLP with hidden width 128 and `tanh` activations.

Latent representation and output. Both branch and trunk produce a 256-dimensional latent representation, interpreted as two 128-dimensional blocks: one for temperature and one for humidity. For each variable, the prediction is obtained by an inner product between the branch and trunk embeddings plus a learned scalar bias.

[Figure placeholder: Architecture diagram of the final DeepONet v0.5.3: branch input = 13 temperature + 13 humidity + 13 mask, trunk input = normalized (x,y), both 6-layer MLPs with width 128, dual output for temperature and humidity.]

Figure 3: Final ThermoDT architecture (v0.5.3). The branch encodes the room state together with explicit missingness information; the trunk encodes the spatial query coordinate. Their per-variable latent blocks are combined to predict temperature and humidity at arbitrary room locations.

4.6 Masking Strategy

Missing sensors are represented in two ways:

(1) the missing sensor values are filled with zero after normalization,
(2) the explicit binary mask tells the network which sensor values are observed.

This prevents ambiguity between genuinely missing values and normalized values that happen to be near zero.

During training, the model sees random masking patterns with varying numbers of masked sensors. This exposes the operator to a broad range of sparse deployment scenarios and makes the final model more robust at evaluation time.

4.7 Training Objective

The final model is trained jointly on temperature and humidity using a supervised loss over masked sensor reconstruction targets. Optimization uses Adam with early stopping on blocked validation loss. The selected checkpoint is the one with the best validation performance.

4.8 Transfer Modes

For Lab 1 → Lab 2 transfer we evaluate three practically relevant modes:

(1) **Zero-shot:** no parameter updates on Lab 2.
(2) **Head-only:** only the output head/bias parameters are updated on a small Lab 2 adaptation split.
(3) **Full fine-tune:** all model parameters are adapted on the Lab 2 adaptation split with a smaller learning rate.

We also tested a partial fine-tune regime, but it underperformed head-only and fullfine-tuning in the current setup, so it is not part of the main final comparison.

4.9 Thermal Comfort Mapping

The final application of ThermoDT is not limited to reconstructing temperature and humidity fields. The reconstructed thermo-hygrometric field is used as input to a thermal comfort model that estimates comfort conditions at arbitrary spatial coordinates within the room. In practice, the DeepONet is queried on a dense set of

Unpublished working draft. Not for distribution.

coordinates, producing spatially continuous estimates of temperature and relative humidity. These predicted fields are then passed to a downstream thermal comfort model to generate a comfort map over the room.

This step is central to the practical value of the project. Temperature and humidity fields are intermediate outputs; the final decision-relevant quantity is the resulting thermal comfort distribution. This allows us to identify regions of the room that are thermally favorable or unfavorable even when no physical sensor is present at those locations.

[Placeholder: describe the specific thermal comfort formulation once finalized. For example, if Justin's model uses PMV/PPD or another comfort proxy, define the equations, required inputs, assumptions, and output interpretation here.]

5 Preliminary Results

5.1 Within-Room Reconstruction in Lab 1

The joint temperature+humidity DeepONet performs strongly under masking and retains good reconstruction quality well beyond the easy one-missing-sensor case. In particular, it maintains strong performance through moderate and even severe sparsity before degrading only at the most extreme masking levels.

Table 1: Example Lab 1 masking performance from the joint temperature+humidity model. Temperature RMSE is in C and humidity RMSE is in %RH. Values shown here are representative of the current benchmark.

Masked	T RMSE	T R^2	RH RMSE	RH R^2
1	0.097	0.962	0.42	0.984
3	0.107	0.956	0.47	0.981
6	0.133	0.945	0.57	0.973
9	0.218	0.866	0.99	0.924
12	0.510	0.365	2.83	0.448

5.2 Minimal Sensor Sets

Exhaustive and greedy subset analysis on Lab 1 shows that a very small number of strategically placed sensors can preserve most of the full-sensor performance. This is one of the strongest practical findings of the project because it connects the model directly to sensor placement and deployment design.

Table 2: Representative minimal active-set results from Lab 1. Composite score is the average of temperature and humidity R^2.

Active set	Score	T R^2	RH R^2
Best 3-sensor set	0.944	0.927	0.962
Best 2-sensor set	0.933	0.916	0.951
Best 1-sensor set	0.839	0.814	0.864

5.3 Architecture Comparison

A compact architecture benchmark showed that the final deeper snapshot MLP (v0.5.3) was one of the strongest models tested while remaining simpler than the LSTM-based alternative. Wider and more complex models did not provide a compelling enough gain to justify adopting them as the main final architecture.

Table 3: Architecture benchmark summary. Mean R^2 is averaged across the comparison protocols used during Step 5.

Model	Main change	Mean R^2
v0.5.1	compact baseline	0.9195
v0.5.2	wider MLP	0.9227
v0.5.3	deeper MLP	0.9301
v0.5.4	short LSTM branch	0.9310

5.4 Cross-Room Transfer: Lab 1 → Lab 2

Cross-room transfer is the central result of the second half of the project. A model trained only on Lab 1 transfers meaningfully to Lab 2 zero-shot, improves further with head-only adaptation, and reaches near-reference performance after full fine-tuning.

[Figure placeholder: Unified transfer overview bar chart comparing LAB1 reference, LAB2 zero-shot, LAB2 head-only, and LAB2 full fine-tune across: all 13 active, top-5 subsets for k=5, top-5 subsets for k=3, and top-5 subsets for k=2.]

Figure 4: Unified transfer overview. The common benchmark consists of all 13 active sensors and the top-5 sparse subsets for $k = 5$, $k = 3$, and $k = 2$. Zero-shot establishes the raw transfer baseline; head-only provides the strongest lightweight adaptation; full fine-tuning yields the strongest final performance.

[Figure placeholder: All-13 per-sensor transfer map for temperature comparing LAB1 reference, LAB2 zero-shot, LAB2 head-only, and LAB2 full fine-tune.]

Figure 5: Per-sensor temperature transfer map under the all-13 regime. The map visualizes where zero-shot transfer is weakest and how head-only and full fine-tuning reduce these spatial failure modes in Lab 2.

5.5 What We Learned from Phase 2

The Phase 2 transfer experiments clarified that not all adaptation modes are equally useful. In the current setup:

Unpublished working draft. Not for distribution.

[Figure placeholder: All-13 per-sensor transfer map for humidity comparing LAB1 reference, LAB2 zero-shot, LAB2 head-only, and LAB2 full fine-tune.]

Figure 6: Per-sensor humidity transfer map under the all-13 regime. The spatial pattern complements the temperature map and highlights that humidity transfer is generally harder, but still improves substantially with adaptation.

[Figure placeholder: Unified subset comparison for top-5 k=5 subsets across LAB1 reference, LAB2 zero-shot, LAB2 head-only, and LAB2 full fine-tune.]

Figure 7: Top-5 subset comparison for $k = 5$. These are the best sparse subsets identified from the unified benchmark and re-evaluated under all transfer modes.

[Figure placeholder: Unified subset comparison for top-5 k=3 subsets across LAB1 reference, LAB2 zero-shot, LAB2 head-only, and LAB2 full fine-tune.]

Figure 8: Top-5 subset comparison for $k = 3$. This figure directly supports the claim that strong transfer is possible even under very sparse sensing.

[Figure placeholder: Unified subset comparison for top-5 k=2 subsets across LAB1 reference, LAB2 zero-shot, LAB2 head-only, and LAB2 full fine-tune.]

Figure 9: Top-5 subset comparison for $k = 2$. This is the most aggressive sparse transfer regime we currently include in the final benchmark.

- **Zero-shot** is already surprisingly strong and establishes the transfer baseline.
- **Head-only** is consistently better than zero-shot and offers a low-cost adaptation strategy.
- **Full fine-tune** is the strongest overall mode and nearly closes the gap to the Lab 1 in-domain reference.

5.6 Application: Spatial Thermal Comfort Assessment

The final application-facing output of the proposed digital twin is a spatial thermal comfort map rather than a temperature or humidity map alone. After reconstructing the temperature and humidity fields with ThermoDT, we evaluate the thermal comfort model on a dense grid of room coordinates. This produces a continuous map of predicted comfort conditions across the space.

[Figure placeholder: insert figure showing temperature field, humidity field, and resulting thermal comfort field for one representative room state. Optionally compare Lab 1 and Lab 2 or compare dense-sensor vs sparse-sensor input conditions.]

Figure 10: TBD

This result is important because it connects the machine learning model to the practical building objective. A sparse wireless sensor network, together with the learned operator, can support comfort-aware monitoring at unsensed locations and help identify spatial regions where occupants may systematically experience discomfort.

6 Experimental Validation Plan

The remaining work is organized around a focused, feasible validation plan rather than a broad set of loosely connected experiments.

6.1 Experiment Group 1: Freeze the Final Within-Room Story

Goal. Finalize the within-room evidence for the selected v0.5.3 model.

Experiments.

(1) Finalize the blocked-split within-room benchmark on Lab 1.
(2) Reconfirm masking robustness on the selected evaluation regimes.
(3) Consolidate the minimal sensor set results and subset rankings for the final report.

Baselines.

- nearest-neighbor interpolation,
- simple spatial interpolation baseline,
- compact DeepONet baseline from earlier iterations.

Metrics. RMSE, MAE, and R^2 for both temperature and humidity.

6.2 Experiment Group 2: Finalize the Transfer Story

Goal. Produce a clean and defensible Lab 1 → Lab 2 transfer benchmark.

Experiments.

(1) Evaluate zero-shot, head-only, and full fine-tune on a common benchmark containing all 13 active sensors and the top-5 subsets for $k = 5$, $k = 3$, and $k = 2$.
(2) Generate per-sensor transfer maps for all-13 active sensors.
(3) Compare sparse subset behavior across the three main transfer modes.

Metrics. RMSE, MAE, and R^2 for temperature and humidity, plus per-sensor R^2 maps.

Unpublished working draft. Not for distribution.

6.3 Experiment Group 3: Thermal Comfort Interpretation

Goal. Connect the reconstructed spatial fields to a building-relevant application.

Experiments.

(1) Select representative reconstructed fields from Lab 1 and Lab 2.
(2) Compute and visualize thermal comfort indicators on top of predicted temperature/humidity fields.
(3) Use these results as the application-facing demonstration of the value of spatial thermal reconstruction.

Metrics. Thermal comfort map quality will be assessed descriptively and, where possible, by consistency with the reconstructed spatial gradients.

6.4 Experiments We Are Not Prioritizing

We are most likely not making temporal prediction, PINN ablations, or uncertainty-guided placement the main focus of the final report. These directions were useful exploratory work, but they would stretch the scope too far relative to the time remaining and the strength of the current core results.

7 Week-by-Week Execution Plan

The plan below covers the remaining three weeks and is designed around what we can realistically finish: final experiments, final writing, and final presentation materials.

Demo day artifacts. The demonstration will likely consist of:

(1) an updated `progress.html` page summarizing the modelling pipeline and key results,
(2) a poster or slide deck showing the final architecture, minimal sensor set findings, and transfer results,
(3) a small visual demonstration of how reconstruction changes under sparse sensor subsets.

Risk mitigation.

- The final paper is centered on results we already have, not on speculative new model directions.
- Writing and figure production will proceed in parallel rather than waiting for all experiments to conclude.
- Exploratory branches such as temporal prediction, PINNs, and uncertainty quantification are treated as optional secondary material rather than required milestones.

8 Conclusion

We have presented ThermoDT, a DeepONet-based thermal digital twin for reconstructing continuous indoor temperature and humidity fields from sparse wireless sensor networks. The project has now converged on a coherent core story: a final sparse-reconstruction model, a principled minimal-sensor analysis, and a strong cross-room transfer result from Lab 1 to Lab 2. Current evidence shows that the selected model reconstructs Lab 1 robustly under severe masking, retains strong performance with very small active sensor sets, and transfers meaningfully across rooms. The remaining work focuses on finalizing this story, tightening the evaluation, and connecting the spatial predictions to thermal comfort interpretation for the final report and demonstration.

Acknowledgments

The authors thank the CMU 12-770 providing real world project opportunity .

References

[1] Juan Botero-Valencia, Calderon Ferney, Alejandro Valencia-Arias, Juan Espinosa-Garcia, and Mario Mejia-Herrera. 2023. Exploring Spatial Patterns in Sensor Data for Humidity, Temperature, and RSSI Measurements. *Data* 8, 6 (2023), 97. doi:10.3390/data8060097
[2] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. 2021. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence* 3, 3 (2021), 218–229. doi:10.1038/s42256-021-00302-5

Unpublished working draft. Not for distribution.

Table 4: Week-by-week plan for the remainder of the project.

Week / Dates	Milestone / Deliverable	Owner(s)	Dependency
Week 1 Apr 8–14	**Freeze the final modelling story.** Finalize the selected model description (v0.5.3), complete the unified Lab 1/Lab 2 transfer benchmark, generate the final figures for sparse subsets and transfer maps, and confirm which results will appear in the paper and website.	David: final model/evaluation runs; Nueraili: figure cleanup and website integration; Justin: finalize thermal comfort model integration and generate thermal comfort maps from the final DeepONet outputs	Existing trained checkpoints and unified evaluation scripts
Week 2 Apr 15–21	**Write the paper around the final results.** Complete the first full draft of Introduction, Related Work, Data, Methodology, and Results. Add the thermal comfort interpretation section and connect it to the spatial reconstruction results. Freeze the main tables, figures, and captions.	Nueraili: Introduction and Related Work; David: Methods and Results; Justin: comfort interpretation and application section	Final figures and frozen metrics
Week 3 Apr 22–28	**Final polish, poster, and demo artifacts.** Revise the full report, finalize the progress webpage, prepare the final poster or slide deck, and build a simple demonstration artifact centered on sparse reconstruction and Lab 1 → Lab 2 transferability.	All team members	Completed paper draft and final visuals

Unpublished working draft.
Not for distribution.